Further Strategies — problem 14.49

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A took the numbers $x_1, \ldots, x_5$, found their pairwise sums $a_1, \ldots, a_{10}$, and challenged B to reconstruct $x_1, \ldots x_5$ from $a_1, \ldots, a_{10}$. Can B succeed ?

Remark : For $n = 2^k$ this is not always possible. For instance, the quadruples $(0, 1, 2, 4)$ and $(-0.5, 1.5, 2.5, 3.5)$ give the same sixtuple $(1, 2, 3, 4, 5, 6)$.